

LogHarbor

Legacy Log Investigation & UTC Reporting

Independent software project | Owner: Ismail Habib | Created September 2026

The screenshot displays the LogHarbor web application interface. On the left is a dark sidebar with the 'LogHarbor MODERNIZATION TOOLKIT' logo and navigation links: 'Log investigation' (active), 'Export reports', and 'C++ analysis engine'. The main content area has a header 'Workspace / Operational diagnostics' and a status indicator 'Local workspace · C++17'. Below this is a section titled 'MAKE OLDER APPLICATION LOGS USEFUL' with the headline 'From scattered log lines to a clear investigation.' and a sub-headline '3 formats one investigation'. The main instruction reads: 'Read application, web-server and Syslog records in one pass. Normalize timestamps, isolate failures and export a report your team can use.' The central form is titled 'Select the log you need to understand' and includes the note 'Use an existing local file or upload UTF-8 log text. Analysis runs entirely on this machine.' It features four input fields: 'LOCAL LOG FILE PATH' (containing '/private/tmp/malt-portfolios-expansion-20260928/14_L'), 'MINIMUM LEVEL' (a dropdown menu set to 'DEBUG'), 'CONTAINS TEXT' (containing 'e.g. timeout'), and 'SYSLOG YEAR' (set to '2026'). An 'Analyze log' button is positioned to the right of these fields. Below the inputs is an 'OR UPLOAD' section with a 'Choose File' button and the text 'No file chosen'. At the bottom of the form, there are four checkmarks: 'Source file preserved', 'UTC normalization', 'Up to 256 MiB via local path', and 'Analysis complete. Source files were not modified.' Below the form are four summary cards: 'LINES INSPECTED' (15, 'One pass · bounded report memory'), 'PARSED EVENTS' (13, '3 formats recognized'), 'MATCHING EVENTS' (13, 'Minimum level: DEBUG'), and 'MALFORMED LINES' (2, 'Skipped with line-number diagnostics'). A footer note at the bottom left of the sidebar states: 'Running locally Your inputs stay on this machine. Independent project by Ismail Habib'.

Actual application capture. Any example records are labeled verification inputs.

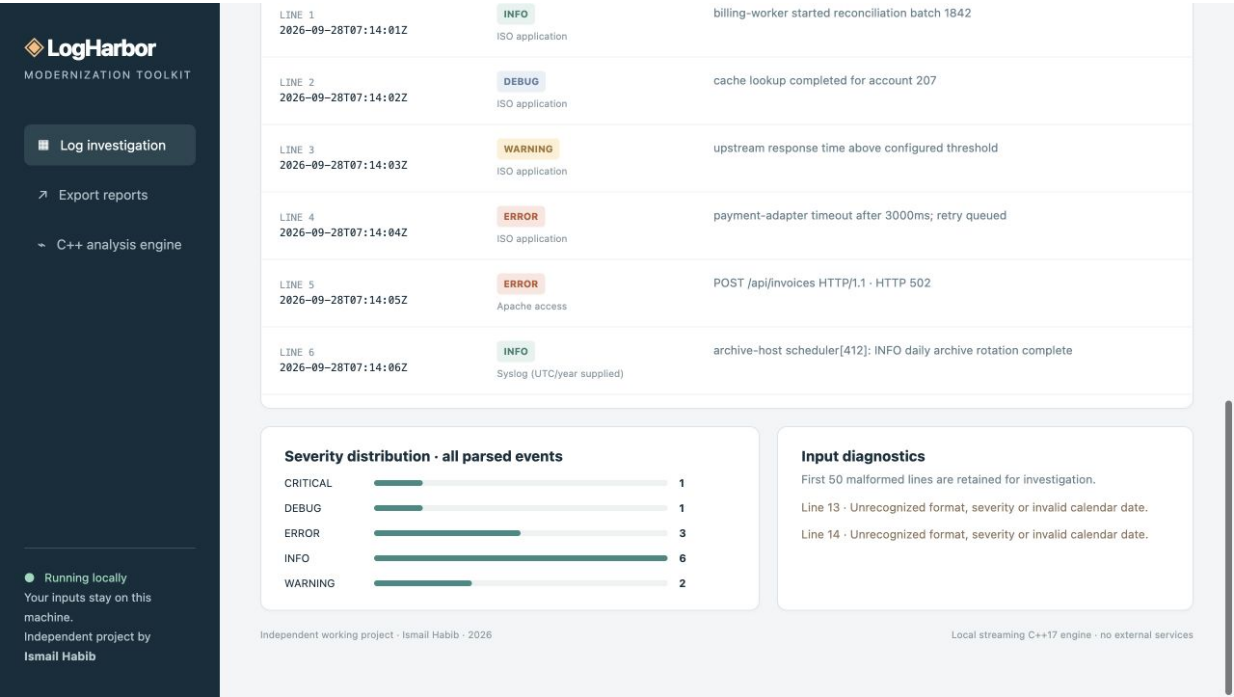
The problem

Older applications and infrastructure often emit incompatible text logs. Mixed time zones, inconsistent severity labels and malformed lines slow incident investigation.

What the project delivers

- Streaming parsing of ISO application, Apache access and classic Syslog text
- UTC normalization, explicit Syslog year and severity mapping
- Case-insensitive message filtering with bounded matching-event output
- Malformed-line diagnostics plus downloadable JSON, CSV and HTML reports

Built to be inspected.



Real parsing output from the labeled synthetic mixed-format fixture, including normalized timestamps and diagnostic totals.

What was verified

Compiled with Apple clang++ in C++17 mode with no warnings. Twelve engine tests and six live HTTP checks passed, including a 25,000-line input with a 500-event output cap. The mixed fixture yielded 13 parsed events and 2 deliberate malformed lines.

How it works

Real UTF-8 log file or upload to streaming C++17 parser to UTC / severity normalization to message filter to investigation / export

Current scope

Independent new modernization-support tool; no historical client migration is claimed. Windows execution and MSVC compilation are not verified. UTF-8 input and three documented formats only. Zone-free timestamps assume UTC; Syslog year is supplied. No multiline stack-trace reconstruction or live tailing. Reports retain the first 500 matches.

Included: source code, setup guide, tests, screenshots, editable case study and Malt copy.
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